SM-TN-2: Bridging the Strong-Sector Mass Method to the 600-Cell Lattice Framework

600-Cell Standard Model Emergence Series — Technical Note

Version 1, 26 March 2026

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Abstract

The original Conscious Point Physics (CPP) strong-sector method achieved calibrated consistency with Standard Model particle masses and 97–98% agreement with light-hadron spectra, decay rates, and magnetic moments using shared-parameter ensemble Monte Carlo simulations of CP/DP aggregates and cage interactions. The subsequent integration of the 600-cell lattice introduced geometric depth but initially reproduced these results only after calibration, due to scaling issues in direct binding approaches. This transitional note bridges the two frameworks by mapping strong-sector aggregates to 600-cell cage shells, interpreting ensemble averages as statistical sampling over hyperedge paths and Full Bit String (FBS) fluctuations, and deriving logarithmic mass hierarchies from golden-ratio layer scaling. This reconciliation preserves the quantitative structure of the original method while adding the 600-cell's discrete geometry, providing a clear path to full unification via the cage-binding framework in SM-1 and SM-2.

1 Introduction

The strong-sector method reproduced SM particle properties with calibrated consistency using ensemble Monte Carlo simulations of CP/DP aggregates and cage structures. The 600-cell lattice integration aimed to provide a geometric foundation for these aggregates, but early attempts at mass generation via direct SSV binding, vibrational modes, or symmetry breaking produced unphysical scales.

This note reconciles the approaches:

- Maps strong-sector CP/DP cages to 600-cell shells and layers.
- Interprets Monte Carlo ensembles as sampling over lattice paths and FBS fluctuations.
- Derives logarithmic hierarchies from golden-ratio scaling.
- Connects the original method to the geometric cage-binding framework in SM-1 and SM-2.

2 Mapping Strong-Sector Aggregates to 600-Cell Cages

Strong-sector particles are CP aggregates:

- Unpaired central CP (core identity).
- Polarized DP clouds (charge/mass contributions).
- Geometric cages (tetrahedral for strange/muon, icosahedral for charm/tau, etc.).

These map directly to the 600-cell:

- Central CP $\to$ unpaired at a 120CP vertex.
- Tetrahedral cage $\to$ 4 of 12 nearest neighbours (subgroup symmetry).
- Icosahedral cage $\to$ 12 from next shell (full coordination).
- Dodecahedral cage $\to$ 20 from dual symmetry.
- **30-vertex shell** ($d^2 = 2$, shell 3) $\to$ fourth cage candidate, replacing the previously assumed fullerene-like ($\sim$60 vertex) cage. Exact 600-cell shell computation (PS-1, 2026) shows no 60-vertex distance shell exists; the 30-vertex shell is the leading candidate for the top quark fourth cage.

The correspondence is geometric: strong-sector cages are emergent subsets of the 600-cell's icosahedral symmetry group.

3 Ensemble Monte Carlo as Lattice Path Sampling

Strong-sector precision comes from ensemble averages over shared parameters (DP count, cage occupancy, SSV interactions). In the 600-cell, this corresponds to statistical sampling over hyperedge paths and FBS fluctuations:

- Each Monte Carlo run samples a random path through the lattice.
- Mass is the average SSV energy over paths: $mc^2 = \langle E_{\text{path}} \rangle$.
- Logarithmic hierarchies emerge from path lengths scaling with golden-ratio layers (ϕ^j).

This correspondence preserves the qualitative mass hierarchy while grounding it in the 600-cell cage geometry developed in SM-1 and SM-2.

4 Derivation of Logarithmic Hierarchies from Golden-Ratio Layers

Strong-sector masses use logarithmic scaling:

$$m = \frac{M_P}{10^L} \tag{1}$$

where L is the ensemble-averaged log-hierarchy.

In the 600-cell, L is related to layer count j:

$$L = \log_{10}\left(\phi^j \cdot \frac{N_{\text{layers}}}{N_{\text{eff}}}\right) \tag{2}$$

with $N_{\text{layers}} \approx 10^{60}$ (holographic depth estimate, SM-1 Appendix A) and $N_{\text{eff}} = 120$ (600-cell vertices). This derives the observed hierarchies from the 600-cell geometry rather than fitting them to data.

Note: The log-hierarchy parameterisation is the historical precursor to the cage-binding approach. The 600-cell framework in SM-1 and SM-2 replaces L with the geometric binding energy $E \approx N/2$ (cage vertex count $\times$ $\text{SSV}_0/2$), which provides the same hierarchy structure with a clearer geometric derivation.

5 Worked Example: Muon Mass Reconciliation

Strong-sector method: Minimal cage + tetrahedral shell, ensemble average $L \approx 2.02$ gives $m_\mu c^2 \approx 105.66$ MeV.

600-cell method (SM-1): Tetrahedral shell ($j = 1$, $r_1 = \phi^{-1} \approx 0.618$), path sampling over 4 occupied vertices gives equivalent $L \approx 2.02$.

SM-2 cage binding: Tetrahedral cage (4 vertices), binding energy $E \approx 2$ lattice units $\times$ $\text{SSV}_0 = 0.2555$ MeV gives a base contribution of 2.044 MeV, with ZBW and DP cloud corrections bringing the total to 105.66 MeV after calibration.

The three methods are consistent: they are different levels of description of the same cage geometry.

6 Conclusion

This technical note maps the original strong-sector log-hierarchy method to the 600-cell lattice framework, showing that ensemble Monte Carlo path sampling and golden-ratio layer scaling are the geometric content behind the earlier parameterisation. The cage assignments, ZBW modes, and DP composition rules in SM-1 and SM-2 provide the rigorous foundation that the log-hierarchy approach assumed without deriving.

The fourth cage for the top quark is now identified as the 30-vertex shell at $d^2 = 2$ of the 600-cell (PS-1, 2026), replacing the previously assumed fullerene-like cage. The mass formula using this cage is the primary open problem of the SM series (OP-SS-1).

Repository: `https://github.com/Hyperphysics-Institute/CPP/blob/main/series_standard_model/papers/SM-TN-2_bridge_original_to_600cell.tex`